phys-MCP Embedded Edge Report

Jetson Orin resource, latency, and control-plane evaluation

Source project: /home/uit/Projects/phys-mcp

Local edge mean	Local edge p95	VDD_IN mean	RAM max
0.457 ms	**0.477 ms**	**5.92 W**	**3114 MB**

Summary: phys-MCP runs on the Jetson Orin as a lightweight embedded control plane. The local edge-control workload completed all 5000 orchestrated runs with sub-millisecond latency. The tegrastats profile records power, RAM, CPU, GPU, and temperature during the workload.

Important scope note: the current workloads exercise the control plane and simulators, not a CUDA-backed model. GR3D/GPU utilization remains 0%, so a future TensorRT, PyTorch CUDA, CUDA kernel, or Ollama backend is needed to prove GPU acceleration.

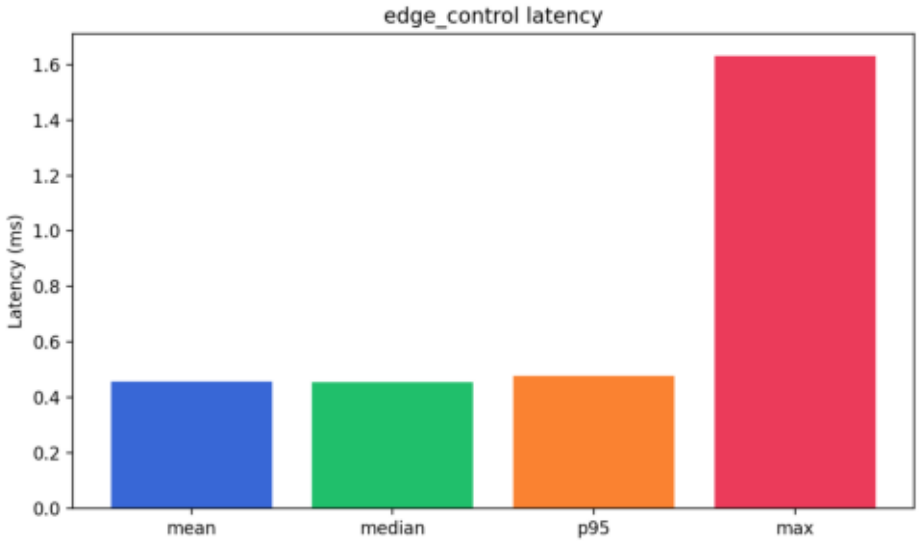
Generated artifacts are stored in evaluation/results/

Local Edge Control Workload

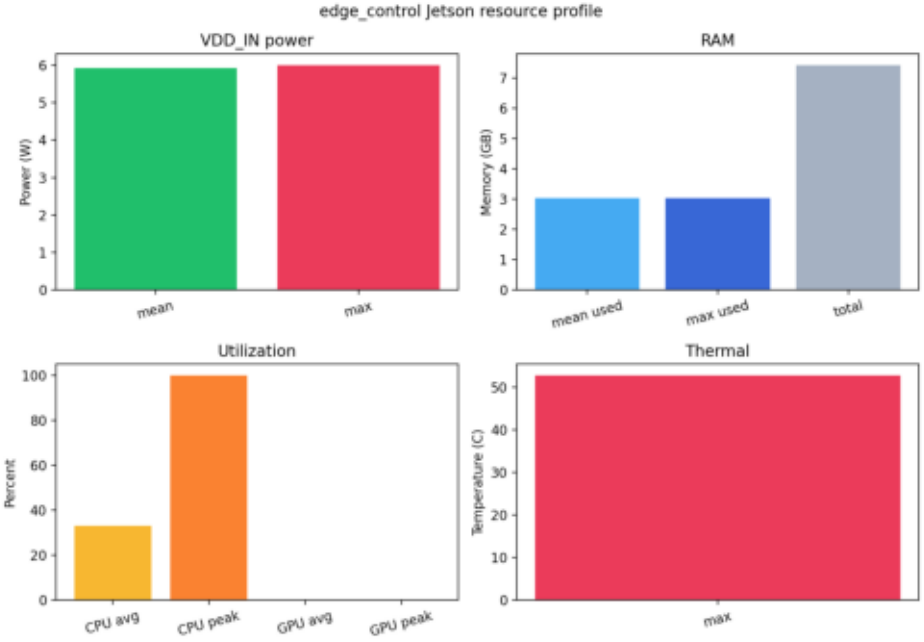
5000 orchestrated edge-control runs under tegrastats sampling

success	mean	p95	RSS delta	GPU
5000/5000	0.457 ms	0.477 ms	28 KB	0%

Latency distribution



Jetson resource profile

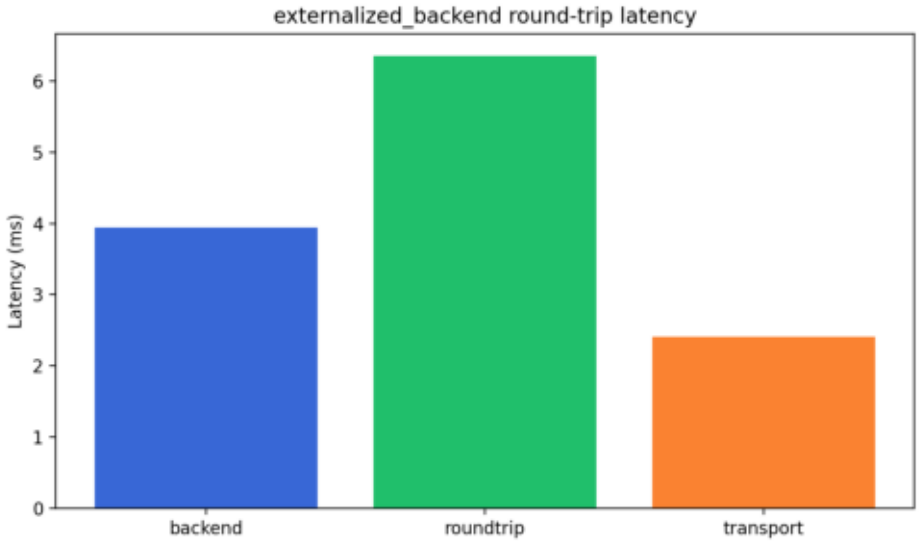


Externalized HTTP Edge Backend

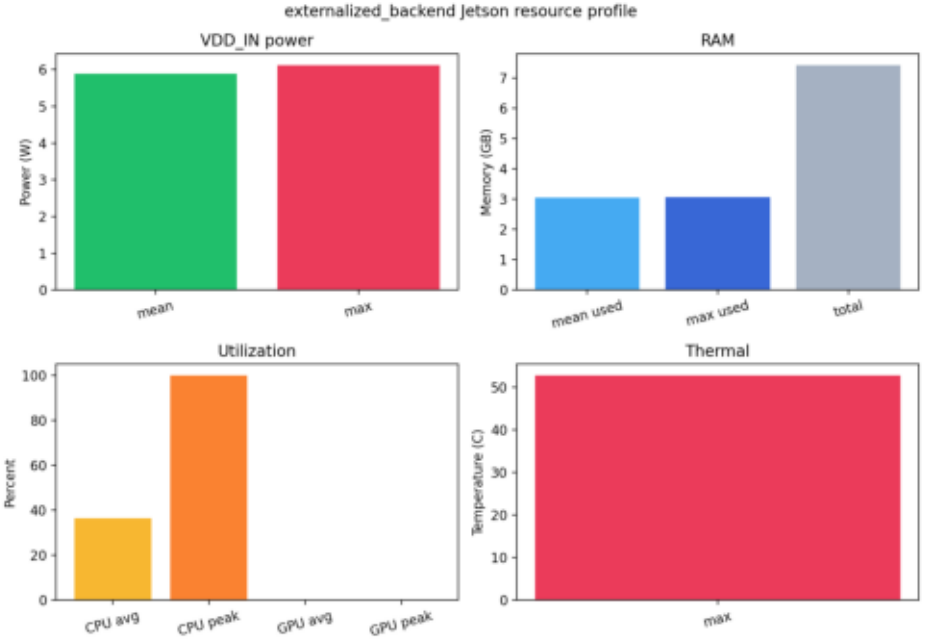
Remote edge adapter path through a local HTTP service boundary

roundtrip mean	backend mean	transport	VDD_IN mean
6.351 ms	3.947 ms	2.405 ms	5.89 W

Round-trip latency



Jetson resource profile



Original vs Orin Timing Comparison

Functional behavior matches the original results; timing shifts are small and remain embedded-friendly.

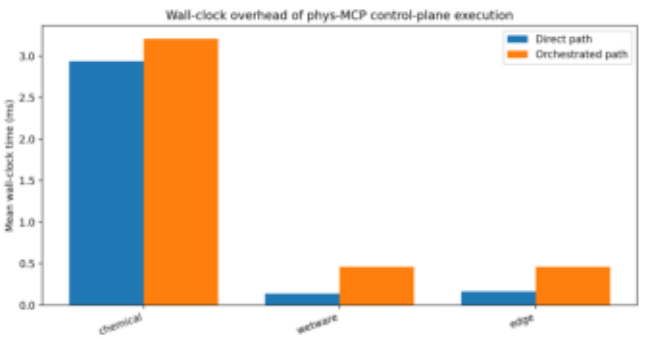
Path	Original mean	Orin mean	Original overhead	Orin overhead
Chemical	2.488 ms	3.208 ms	0.361 ms	0.268 ms
Wetware	0.267 ms	0.460 ms	0.194 ms	0.322 ms
Edge	0.321 ms	0.463 ms	0.189 ms	0.301 ms
HTTP edge	8.961 ms	6.351 ms	5.015 ms	2.405 ms

The Orin is slightly slower for the fastest in-process synthetic paths but stays below 0.5 ms for the local edge control path. The local HTTP edge boundary is faster than the original committed result in this run.

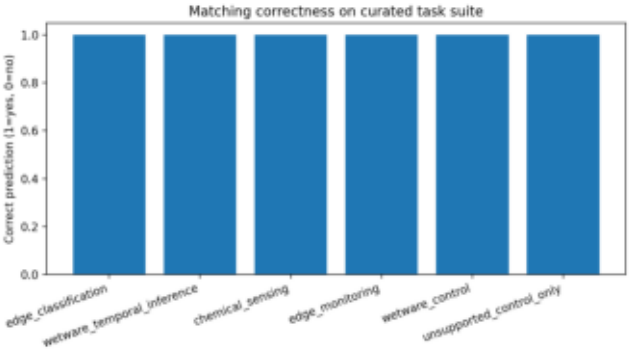
Existing Evaluation Charts

Control overhead, matcher accuracy, and descriptor portability

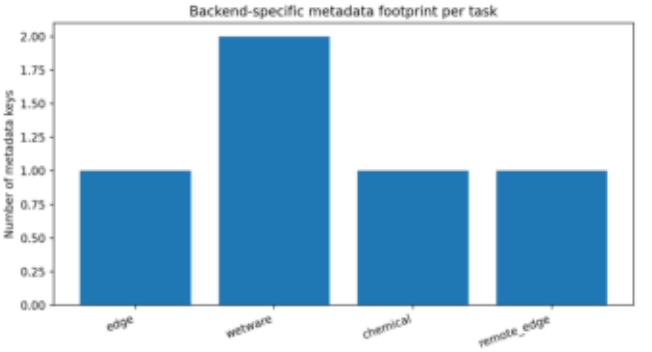
Control-plane overhead



Matcher accuracy



Portability metadata



Matcher evidence: full phys-MCP selector reaches 7/7 on the baseline suite, while simpler modality-only and latency-only selectors miss runtime-sensitive cases.

Portability evidence: descriptor shared-key ratio is 1.0, invocation shared-key ratio is 1.0, and 4/4 executable backend runs succeeded.

Robustness and Integration Evidence

Representative runtime faults are handled by rejection or fallback as expected.

Scenario	Selected backend	Expected?
drifted primary edge avoided	remote-edge-backend	yes
prepare failure with fallback	remote-edge-backend	yes
policy violation rejected	none	yes
stale twin rejected	none	yes
telemetry loss triggers fallback	remote-edge-backend	yes

Cortical Labs simulator path: 3/3 successful runs, 64 channels, 25000 FPS. Current conclusion: embedded control-plane feasibility is supported; GPU acceleration remains future work.